Lecture 25

# Poincaré representations and little groups

**Objectives.** *This lecture reduces the classification of positive-energy irreducible unitary representations of the Poincaré group to little groups. It makes the massive and finite-helicity sectors explicit. Translations are diagonalised by momenta. Lorentz transformations move momenta along orbits. The orbit and an irreducible unitary representation of the stabiliser of one of its points determine the representation up to unitary equivalence.*

By the end of this lecture you will be able to:

- Construct the group $\mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ and its action on momenta.
- Classify the momentum orbits and compute the massive and massless little groups.
- State Mackey's classification with its hypotheses, and separate the finite-helicity, continuous-spin and zero-momentum sectors.
- Verify the massive spin model and name the steps it omits.

## 25.1 The Poincaré group as a semidirect product

Spacetime is $\mathbb{R}^4$ with points $x = (x_0, x_1, x_2, x_3)$, where $x_0$ is time. The *Minkowski pairing* is

$$x \cdot y = x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Lorentz transformations are the linear maps preserving this pairing. The proper orthochronous ones form the group $\mathrm{SO}^+(1,3)$. Adding translations gives the Poincaré group. Quantum symmetries act projectively, and Lecture 13 explains why one then passes to a cover. We prove no lifting theorem. We study unitary representations of the cover built below.

Identify $\mathbb{R}^4$ with the $2 \times 2$ Hermitian matrices by

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \qquad \det \Phi(x) = x \cdot x, \qquad \operatorname{tr} \Phi(x) = 2x_0. \tag{25.1}$$

For $g \in \mathrm{SL}(2,\mathbb{C})$ define the map $\Lambda(g) \colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$\Phi(\Lambda(g)x) = g\, \Phi(x)\, g^*.$$

**Lemma 25.1.** *The map $\Lambda(g)$ is linear, and its entries are real polynomials in the real and imaginary parts of the entries of $g$. Moreover $\Lambda(gg') = \Lambda(g)\Lambda(g')$ and $\Lambda(g)x \cdot \Lambda(g)y = x \cdot y$.*

*Proof.* The matrix $g\Phi(x)g^{*}$ is Hermitian and depends linearly on $x$. For a Hermitian matrix $h = (h_{ij})$, the vector $\Phi^{-1}(h)$ depends linearly on $h$. For example, its time coordinate is $x_0 = \tfrac{1}{2}(h_{11} + h_{22})$. Hence $\Lambda(g)$ is linear. Each entry of $g\Phi(x)g^{*}$ is a sum of products of an entry of $g$, an entry of $\Phi(x)$ and the complex conjugate of an entry of $g$. So the entries of $\Lambda(g)$ are real polynomials in the real and imaginary parts of the entries of $g$.

The second claim follows from $gg'\Phi(x)(gg')^{*} = g\big(g'\Phi(x)g'^{*}\big)g^{*}$. For the third, $\det(g\Phi(x)g^{*}) = |\det g|^{2}\det\Phi(x) = x\cdot x$. Polarisation, $2x\cdot y = (x+y)\cdot(x+y) - x\cdot x - y\cdot y$, gives the pairing. ∎

By Lecture 24, $\Lambda$ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^{+}(1,3)$ with kernel $\{\pm I\}$. The final claim of Proposition 25.3 and the descent statements use this fact. The classification itself does not depend on it.

**Definition 25.2.** The group $G = \mathbb{R}^{4} \rtimes \mathrm{SL}(2,\mathbb{C})$ is the set $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ with the product

$$(a,g)(a',g') = \big(a + \Lambda(g)a',\; gg'\big). \tag{25.2}$$

**Proposition 25.3.** *$G$ is a Lie group of dimension 10. Its identity is $(0,I)$ and $(a,g)^{-1} = \big(-\Lambda(g)^{-1}a,\; g^{-1}\big)$. The translations $A = \{(a,I)\} \cong \mathbb{R}^{4}$ form a closed normal subgroup, and*

$$(0,g)(a,I)(0,g)^{-1} = \big(\Lambda(g)a,\; I\big). \tag{25.3}$$

*The map $(a,g) \mapsto (a,\Lambda(g))$ is a homomorphism onto $\mathbb{R}^{4} \rtimes \mathrm{SO}^{+}(1,3)$ with kernel $\{(0,\pm I)\}$.*

*Proof.* Associativity is the main point. Both bracketings of a triple product have second entry $gg'g''$. Their first entries are

$$a + \Lambda(g)a' + \Lambda(gg')a'' \qquad \text{and} \qquad a + \Lambda(g)\big(a' + \Lambda(g')a''\big).$$

These agree because $\Lambda(g)$ is linear and $\Lambda$ is multiplicative, by Lemma 25.1. The stated inverse is an inverse on both sides, since

$$(a,g)\big(-\Lambda(g)^{-1}a,\, g^{-1}\big) = (a-a,\; I), \qquad \big(-\Lambda(g)^{-1}a,\, g^{-1}\big)(a,g) = \big(-\Lambda(g)^{-1}a + \Lambda(g)^{-1}a,\; I\big).$$

The manifold $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ has dimension $4 + 6$. Multiplication is polynomial in real coordinates by Lemma 25.1. So is inversion, since $g^{-1}$ is the adjugate of $g$. Hence $G$ is a Lie group. The projection $(a,g) \mapsto g$ is a continuous homomorphism with kernel $A$, so $A$ is closed and normal. Next, $(0,g)(a,I) = (\Lambda(g)a, g)$, and multiplying on the right by $(0,g^{-1})$ gives (25.3). The final map respects (25.2), because only $\Lambda(g)$ acts on translations. Its image and kernel come from those of $\Lambda$. ∎

From now on, the Poincaré group means $G$. The central element $z = (0,-I)$ maps to the identity.

## 25.2 Momentum characters and mass shells

Let $U$ be a strongly continuous unitary representation of $G$ on a Hilbert space. On $A$ it gives commuting unitary operators. Their building blocks are characters. Exercise 25.2 proves the next lemma.

**Lemma 25.4.** *Every continuous homomorphism $\mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.*

The sign follows the generator convention of Lecture 15. For the one-dimensional representation $\chi_p$ and the translation generator $X = e_0$, we have $\chi_p(te_0) = e^{-itp_0}$. Thus $A_X = p_0$ in that convention, and $p_0$ is the energy.

A *projection-valued measure* $E$ on $\mathbb{R}^4$ assigns an orthogonal projection $E(S)$ to every Borel set $S \subseteq \mathbb{R}^4$, with $E(\mathbb{R}^4) = I$ and countable additivity in the strong operator topology for disjoint sets. We say that $E$ is *concentrated* on a Borel set $S$ when $E(S) = I$.

We cite the *Stone–Naimark–Ambrose–Godement (SNAG) theorem* for vector groups; see Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2. For a strongly continuous unitary representation of $\mathbb{R}^4$, it gives a unique projection-valued measure $E$ on $\mathbb{R}^4$ with

$$U(a, I) = \int_{\mathbb{R}^4} \chi_p(a)\, dE(p).$$

The *support* of $E$ is the complement of the union of open sets $O$ with $E(O) = 0$. Since $\mathbb{R}^4$ has a countable basis, this union is a countable union of such null sets. Thus the support is the smallest closed set $F$ with $E(F) = I$. We call it the *translation spectrum*.

We use the spectral calculus for projection-valued measures as a cited analytic input (Hall, *Quantum Theory for Mathematicians*, §10.1). It defines the self-adjoint operator

$$P_0 = \int_{\mathbb{R}^4} p_0\, dE(p), \qquad \mathrm{Dom}(P_0) = \left\{ v : \int_{\mathbb{R}^4} p_0^2\, d\langle v, E(p)v\rangle < \infty \right\}.$$

By Stone's theorem with the convention of Lecture 15, this is the generator of $t \mapsto U(te_0, I)$. Positive energy means $P_0 \geq 0$. By the spectral calculus this is equivalent to concentration of $E$ on $\{p_0 \geq 0\}$. Momentum eigenvectors need not exist in the Hilbert space.

Still, a formal computation shows the mechanism. If $U(a, I)\psi = \chi_p(a)\psi$ for all $a$, then (25.3) gives

$$U(a, I)U(0, g)\psi = U(0, g)\, U(\Lambda(g)^{-1}a, I)\psi = \chi_{\Lambda(g)p}(a)\, U(0, g)\psi. \tag{25.4}$$

Here $p \cdot \Lambda(g)^{-1}a = \Lambda(g)p \cdot a$ by the pairing invariance in Lemma 25.1. The same calculation with $E$, together with the uniqueness of $E$ in the SNAG theorem, gives

$$U(0, g)E(S)U(0, g)^{-1} = E(\Lambda(g)S).$$

Hence the translation spectrum is stable under the following action.

**Definition 25.5.** The *mass-shell action* of $\mathrm{SL}(2, \mathbb{C})$ on momentum space $\mathbb{R}^4$ is $g \cdot p = \Lambda(g)p$. For $m, \kappa > 0$ set

$$X_m^{\pm} = \{p \cdot p = m^2,\ \pm p_0 > 0\}, \qquad C^{\pm} = \{p \cdot p = 0,\ \pm p_0 > 0\}, \qquad Y_\kappa = \{p \cdot p = -\kappa^2\}.$$

**Theorem 25.6.** *The orbits of the mass-shell action are exactly the sets $X_m^{\pm}$, $C^{\pm}$, $Y_\kappa$ and $\{0\}$.*

*Proof. Invariants.* The number $p \cdot p$ is invariant, because $\Lambda(g)$ preserves the pairing by Lemma 25.1. Write $\vec{p} = (p_1, p_2, p_3)$, with Euclidean norm $|\vec{p}|$. Suppose $p \cdot p \geq 0$ and $p \neq 0$. Then $p_0^2 \geq |\vec{p}|^2$, so $p_0 \neq 0$. The eigenvalues $\mu_1 \geq \mu_2$ of $\Phi(p)$ satisfy

$$\mu_1 \mu_2 = p \cdot p \geq 0, \qquad \mu_1 + \mu_2 = 2p_0.$$

So $p_0 > 0$ exactly when $\Phi(p)$ is positive semidefinite and nonzero. The map $\Phi \mapsto g\Phi g^*$ preserves both properties, because $v^* g \Phi g^* v = (g^* v)^* \Phi (g^* v)$ and $g$ is invertible. Hence each listed set is a union of orbits. The listed sets partition $\mathbb{R}^4$.

*Transitivity.* Write $\Phi(p) = u \operatorname{diag}(\mu_1, \mu_2)\, u^*$ with $u \in \mathrm{U}(2)$. A scalar factor $e^{i\phi}$ with $e^{2i\phi} \det u = 1$ puts $u$ in $\mathrm{SU}(2)$ and changes nothing. Choose the base points

$$k_m = (m, 0, 0, 0), \qquad k = (1, 0, 0, 1), \qquad \ell_\kappa = (0, 0, 0, \kappa).$$

Their $\Phi$-images are $\operatorname{diag}(m, m)$, $\operatorname{diag}(2, 0)$ and $\operatorname{diag}(\kappa, -\kappa)$. For $s > 0$ let $d_s = \operatorname{diag}(s, s^{-1})$. Then

$$d_s \operatorname{diag}(\alpha, \beta)\, d_s^* = \operatorname{diag}(s^2 \alpha,\ s^{-2} \beta). \tag{25.5}$$

On $X_m^+$ we have $\mu_1, \mu_2 > 0$ and $\mu_1 \mu_2 = m^2$. Take $s^2 = \mu_1 / m$. Then $s^{-2} m = \mu_2$, so $p = \Lambda(u d_s) k_m$. On $C^+$ we have $\mu_1 = 2p_0$ and $\mu_2 = 0$; take $s^2 = \mu_1 / 2$. On $Y_\kappa$ we have $\mu_1 > 0 > \mu_2$ and $\mu_1 \mu_2 = -\kappa^2$; take $s^2 = \mu_1 / \kappa$. Since $\Lambda(g)$ is linear, $X_m^-$ and $C^-$ are the orbits of $-k_m$ and $-k$. Finally 0 is fixed. ∎

By (25.5), $\Lambda(d_s)$ multiplies $p_0 + p_3$ by $s^2$ and $p_0 - p_3$ by $s^{-2}$. It is a boost along the $x_3$-axis.

A subset of $\mathbb{R}^4$ is *locally closed* when it is the intersection of a closed set and an open set. Each orbit is locally closed. For example, $C^+$ is the closed set $\{p \cdot p = 0\}$ intersected with the open set $\{p_0 > 0\}$.

The invariant Borel sets $\{p \cdot p < r\}$ with $r$ rational, together with $\{p \cdot p \geq 0,\ p_0 > 0\}$, $\{p \cdot p \geq 0,\ p_0 < 0\}$ and $\{0\}$, separate the orbits. Local closedness and this separation property form the regularity used below.

## 25.3 Little groups and Mackey's classification

**Definition 25.7.** The *little group* of $p \in \mathbb{R}^4$ is its stabiliser $H_p = \{g \in \mathrm{SL}(2, \mathbb{C}) : \Lambda(g) p = p\}$.

For $b \in \mathbb{C}$ and $|\lambda| = 1$ write

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix}.$$

**Proposition 25.8.** *(a)* $H_{k_m} = \mathrm{SU}(2)$.

*(b)* $H_k = \{n(b) r(\lambda) : b \in \mathbb{C},\ |\lambda| = 1\}$, *and* $r(\lambda) n(b) r(\lambda)^{-1} = n(\lambda^2 b)$.

*(c)* $\Lambda(r(e^{-i\theta/2}))$ *rotates the* $(x_1, x_2)$*-plane by the angle* $+\theta$ *and fixes* $x_0$ *and* $x_3$.

*(d)* $H_{\ell_\kappa} = \{g : g \sigma_3 g^* = \sigma_3\} = \mathrm{SU}(1, 1)$, *where* $\sigma_3 = \operatorname{diag}(1, -1)$. *Also* $H_0 = \mathrm{SL}(2, \mathbb{C})$.

*Proof.* (a) The condition is $g(mI)g^{*} = mI$, that is, $gg^{*} = I$.

(b) Since $\Phi(k) = 2e_1e_1^{*}$, the condition is $vv^{*} = e_1e_1^{*}$ for $v = ge_1$. In entries,

$$\begin{pmatrix} |v_1|^2 & v_1\bar{v}_2 \\ v_2\bar{v}_1 & |v_2|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $v_2 = 0$ and $v_1 = \lambda$ with $|\lambda| = 1$. Since $\det g = 1$,

$$g = \begin{pmatrix} \lambda & w \\ 0 & \bar{\lambda} \end{pmatrix} = n(\lambda w)\, r(\lambda).$$

Every such $g$ fixes $k$. The conjugation formula is a product of three matrices:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar{\lambda} \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \bar{\lambda} & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 & \lambda^2 b \\ 0 & 1 \end{pmatrix}.$$

(c) Conjugation by $r(\lambda)$ fixes the diagonal of $\Phi(p)$. It multiplies the entry $p_1 + ip_2$ by $\bar{\lambda}^2$, which is $e^{i\theta}$ for $\lambda = e^{-i\theta/2}$.

(d) This restates the definition, since $\Phi(\ell_\kappa) = \kappa\sigma_3$. ∎

Write $N = \{n(b)\} \cong \mathbb{C}$ and $T = \{r(\lambda)\} \cong \mathrm{U}(1)$. By the factorisation in Proposition 25.8(b), $H_k = N \rtimes T$. The map $n(b)r(\lambda) \mapsto (b, \lambda^2)$ is two-to-one onto the group of orientation-preserving Euclidean motions of the plane. So we call $N$ the *little-group translations*. The maps $\Lambda(n(b))$ are Lorentz transformations fixing $k$, not spacetime translations.

We have $r(e^{-i\theta/2}) = \exp(\theta X_3)$ with $X_3 = -i\sigma_3/2$. By the rotation formula in Proposition 25.8(c), this generates positive-angle rotations about $x_3$, as in Lecture 24.

The next theorem uses unitary induction from Lecture 17.

**Theorem 25.9** (Mackey, cited). *Let $G = \mathbb{R}^n \rtimes H$, where $H$ is a second-countable Lie group acting linearly and continuously on $\mathbb{R}^n$. For $p \in (\mathbb{R}^n)^{*}$ put $\chi_p(a) = e^{-ip(a)}$. Let $H$ act on these momenta so that $\chi_{h\cdot p}(a) = \chi_p(h^{-1}\cdot a)$. Assume the action is* regular*: orbits are locally closed, and countably many invariant Borel sets separate them. Let $H_p$ be the stabiliser of $p$. For a strongly continuous unitary representation $U$ of $G$, let $E$ be the projection-valued measure on $(\mathbb{R}^n)^{*}$ that the SNAG theorem attaches to the translations $a \mapsto U(a, e)$.*

*(a) If $U$ is irreducible, then $E$ is concentrated on one orbit $\Omega$.*

*(b) Let $p \in \Omega$ and let $\sigma$ be a strongly continuous irreducible unitary representation of $H_p$. Put $(\chi_p \otimes \sigma)(a, h) = \chi_p(a)\sigma(h)$. Then $\mathrm{Ind}_{\mathbb{R}^n \rtimes H_p}^{G}(\chi_p \otimes \sigma)$ is irreducible, and its measure $E$ is concentrated on $\Omega$.*

*(c) Every strongly continuous irreducible unitary representation whose measure $E$ is concentrated on $\Omega$ is unitarily equivalent to one of the induced representations in (b). For a fixed $p \in \Omega$, two of these induced representations are unitarily equivalent exactly when their representations $\sigma$ are unitarily equivalent.*

This is Mackey's method for semidirect products, as presented in A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. For our $G$ it is Wigner's classification of 1939. For our spacetime group, the Minkowski pairing identifies the functionals $p \in (\mathbb{R}^4)^{*}$ of the theorem with the vectors $p$ used above. The remarks after the orbit theorem, Theorem 25.6, verify regularity. The dual action is $g \cdot p = \Lambda(g)p$ by (25.4).

**Corollary 25.10.** *(a) For $m > 0$, up to unitary equivalence, the strongly continuous irreducible unitary representations of $G$ with $E$ concentrated on $X_m^+$ are labelled by $n \geq 0$. The label is the* SU(2)*-representation $V_n$, of dimension $n+1$ and spin $n/2$.*

*(b) Up to unitary equivalence, those with $E$ concentrated on $C^+$ are labelled by the equivalence classes of strongly continuous irreducible unitary representations of $H_k = N \rtimes T$.*

*Proof.* Combine Mackey's theorem, Theorem 25.9, with the little groups computed in Proposition 25.8. In the massive case, strongly continuous irreducible unitary representations of the compact group SU(2) are finite-dimensional by Lecture 10. Hence they are the $V_n$ of Lecture 8. ∎

## 25.4 Massless sectors: helicity and continuous spin

**Proposition 25.11.** *Let $\sigma$ be a finite-dimensional continuous unitary representation of $H_k$ on $V$. Then $\sigma(n(b)) = I$ for all $b$. If $\sigma$ is irreducible, then $\sigma(n(b)r(\lambda)) = \lambda^\nu$ for a unique $\nu \in \mathbb{Z}$, and*

$$\sigma(\exp \theta X_3) = e^{-i\theta\nu/2}.$$

*Proof.* The operators $\sigma(n(b))$ are unitary and commute. So $V$ is the direct sum of their joint eigenspaces. On each, $b \mapsto \sigma(n(b))$ is a continuous character of $\mathbb{C} \cong \mathbb{R}^2$. As in the character lemma, Lemma 25.4, it has the form $b \mapsto e^{-i\,\mathrm{Re}(\bar\zeta b)}$ for a unique $\zeta \in \mathbb{C}$. Let $V_\zeta$ be the joint eigenspace for $\zeta$. Let $S$ be the finite set of $\zeta$ with $V_\zeta \neq 0$.

This is the key step. By the conjugation formula in Proposition 25.8(b), $n(b)r(\lambda) = r(\lambda)n(\bar\lambda^2 b)$. Take $v \in V_\zeta$. Then

$$\sigma(n(b))\sigma(r(\lambda))v = \sigma(r(\lambda))\sigma(n(\bar\lambda^2 b))v = e^{-i\,\mathrm{Re}(\overline{\lambda^2\zeta}\, b)}\,\sigma(r(\lambda))v.$$

So $\sigma(r(\lambda))V_\zeta = V_{\lambda^2\zeta}$. Every unit complex number is some $\lambda^2$. If some $\zeta \neq 0$ lay in $S$, the whole circle $|\zeta'| = |\zeta|$ would lie in $S$. This contradicts finiteness. So $S = \{0\}$, and $N$ acts trivially.

Then $\sigma$ factors through $n(b)r(\lambda) \mapsto \lambda$. This is a homomorphism onto U(1), since $n(b)r(\lambda)n(b')r(\lambda') = n(b + \lambda^2 b')r(\lambda\lambda')$. The operators $\sigma(r(\lambda))$ commute. If $\sigma$ is irreducible, a common eigenline is invariant, so $\dim V = 1$. By the one-variable character argument of Exercise 25.2(a), $\theta \mapsto \sigma(r(e^{i\theta}))$ equals $e^{i\nu\theta}$ for some real $\nu$. Periodicity in $\theta$ forces $\nu \in \mathbb{Z}$. Finally $\exp \theta X_3 = r(e^{-i\theta/2})$ maps to $e^{-i\nu\theta/2}$. ∎

Compare with the generator convention $\Pi(\exp \theta X) = \exp(-i\theta A_X)$ of Lecture 15. The self-adjoint generator of rotations about the momentum direction is $A_{X_3} = \nu/2$. This number is the *helicity*.

The *finite-helicity* massless representations are the representations induced from $\chi_k \otimes \sigma$, where $\sigma$ is a finite-dimensional continuous irreducible unitary representation of $H_k$. By Proposition 25.11, these are the representations $\mathrm{Ind}(\chi_k \otimes \sigma_\nu)$ with $\nu \in \mathbb{Z}$ and $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$. Each has the single helicity $\nu/2$. In these representations the little-group translations act trivially.

**Remark 25.12.** Mackey's theorem, Theorem 25.9, also applies to $H_k = N \rtimes T$. The $T$-orbits on the characters of $N$ are $\{0\}$ and the circles $|\zeta| = \rho > 0$. These orbits are closed. The sets $\{|\zeta| < r\}$ with $r \in \mathbb{Q}$, together with $\{0\}$, separate them. Hence the action is regular.

The orbit $\{0\}$ gives the $\sigma_\nu$. A circle has stabiliser $\{r(\pm 1)\}$ with two characters. These give infinite-dimensional irreducible representations of $H_k$ that are nontrivial on $N$ (Exercise 25.7). Inducing them to $G$ gives the *continuous-spin* representations. Their spectral measure is concentrated on $C^+$, so their energy is positive. Positive energy alone does not exclude them. They form a sector separate from finite helicity.

Now suppose a nonzero representation has $E$ concentrated on $C^+$. Its closed support meets $C^+$ and is Lorentz-invariant by covariance of $E$. Transitivity then puts all of $C^+$ in the support. Closedness adds its limit point 0. So concentration on an orbit differs from equality of closed supports.

The orbit $\{0\}$ is also separate. Translations act trivially there. Any strongly continuous irreducible unitary representation of $\mathrm{SL}(2,\mathbb{C})$ gives a representation of this zero-momentum sector by composition with the projection $G \to \mathrm{SL}(2,\mathbb{C})$. The trivial one is the usual vacuum choice. Positive energy alone does not select it.

Positive energy excludes $X_m^-$ and $C^-$, where $p_0 < 0$. It also excludes $Y_\kappa$. Indeed, the set $S = Y_\kappa \cap \{p_0 < 0\}$ is nonempty and relatively open. By transitivity, the Lorentz translates of $S$ cover $Y_\kappa$. By second countability, countably many of these translates already cover $Y_\kappa$. Positive energy gives $E(S) = 0$, since $S \subset \{p_0 < 0\}$. Then covariance of $E$ makes every translate of $S$ an $E$-null set. Disjointising this countable cover and using countable additivity would give $E(Y_\kappa) = 0$. This contradicts concentration of $E$ on that orbit in a nonzero representation.

## 25.5 The massive induced model

Fix $m > 0$ and $n \geq 0$. Let $D_n$ be the irreducible unitary representation of $\mathrm{SU}(2)$ on $V_n$. Put $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$. Define a measure $\mu_m$ on $X_m^+$ by

$$\int_{X_m^+} f \, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p}) \, \frac{d^3\vec{p}}{2\,\omega(\vec{p})}.$$

For $p \in X_m^+$ the matrix $\Phi(p)$ is positive definite. Let $h_p = (\Phi(p)/m)^{1/2}$ be its positive square root. Then $\det h_p = 1$ and $h_p(mI)h_p^* = \Phi(p)$, so $\Lambda(h_p)k_m = p$. The *Wigner rotation* is

$$W(g,p) = h_p^{-1}\, g\, h_{\Lambda(g)^{-1}p}.$$

The model space is $\mathcal{H}_{m,n} = L^2(X_m^+, \mu_m; V_n)$. The operators are

$$(U(a,g)\psi)(p) = e^{-ip\cdot a}\, D_n\big(W(g,p)\big)\, \psi(\Lambda(g)^{-1}p). \tag{25.6}$$

**Proposition 25.13.** *(a)* $W(g,p) \in \mathrm{SU}(2)$.
*(b)* $W(gg',p) = W(g,p)\,W(g',\Lambda(g)^{-1}p)$.
*(c)* $U((a,g)(a',g')) = U(a,g)\,U(a',g')$.
*(d) If $\mu_m$ is $\Lambda$-invariant, then each $U(a,g)$ is unitary.*
*(e) If $g \in \mathrm{SU}(2)$, then $W(g,p) = g$.*

*Proof.* (a) Put $q = \Lambda(g)^{-1}p$. Then $h_q$ maps $k_m$ to $q$, $g$ maps $q$ to $p$, and $h_p^{-1}$ maps $p$ to $k_m$. So $W(g,p) \in H_{k_m} = \mathrm{SU}(2)$.

(b) With $q$ as above, $W(g,p)W(g',q) = h_p^{-1} g h_q \, h_q^{-1} g' h_{\Lambda(g')^{-1}q}$. The middle factors cancel, and $\Lambda(g')^{-1}q = \Lambda(gg')^{-1}p$.

(c) Apply (25.6) twice:

$$U(a,g)U(a',g')\psi(p) = e^{-ip\cdot a}\, e^{-iq\cdot a'}\, D_n(W(g,p))\, D_n(W(g',q))\, \psi(\Lambda(g')^{-1}q).$$

By the pairing invariance in Lemma 25.1, $q \cdot a' = p \cdot \Lambda(g)a'$, so the phase is $e^{-ip\cdot(a+\Lambda(g)a')}$. By the cocycle identity (b), the operator is $D_n(W(gg',p))$. The argument is $\Lambda(gg')^{-1}p$. By (25.2) this is $U(a+\Lambda(g)a', gg')\psi(p)$.

(d) The phase has modulus one and $D_n$ is unitary. Therefore

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2 \, d\mu_m(p) = \|\psi\|^2.$$

By the homomorphism property (c), $U(a,g)$ has inverse $U((a,g)^{-1})$.

(e) For unitary $g$, the matrix $gh_qg^{-1}$ is positive definite, and its square is $g\Phi(q)g^*/m = \Phi(p)/m$. By uniqueness of the positive square root, $gh_qg^{-1} = h_p$. Hence $W(g,p) = g$. ∎

By the rotation case of the Wigner rotation, Proposition 25.13(e), rotations act by $(U(0,u)\psi)(p) = D_n(u)\psi(\Lambda(u)^{-1}p)$. At the rest momentum $k_m$ the little group acts through $D_n$, which is spin $n/2$. The central element acts by

$$U(z) = D_n(-I) = (-1)^n I.$$

Hence half-integer spin does not descend to $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$. Time translation by $t$ acts by multiplication by $e^{-itp_0}$. Its self-adjoint generator $P_0$ is multiplication by $p_0$, with domain

$$\mathrm{Dom}(P_0) = \{\psi \in \mathcal{H}_{m,n} : p_0\psi \in \mathcal{H}_{m,n}\}.$$

Since $p_0 \geq m$ on $X_m^+$, we get $P_0 \geq m$. The energy is positive.

Three steps complete the identification with a unitary induced representation. The measure and intertwining calculations are assigned to the sheet. Continuity is explained here.

1. *Invariance of $\mu_m$.* Exercise 25.4 writes $g = u\, d_s\, v$ with $u, v \in \mathrm{SU}(2)$. Rotations fix $p_0$ and act orthogonally on $\vec{p}$. The boost $d_s$ needs a Jacobian computation.
2. *Strong continuity.* For continuous compactly supported $\psi$, the formula is jointly continuous in $(a,g,p)$. The positive square root $h_p$ is continuous by matrix functional calculus. For $(a,g)$ in a compact neighbourhood, transformed supports lie in one compact set and the integrands are uniformly bounded there. Dominated convergence gives norm continuity. Density and unitarity extend it to all $\psi$.
3. *Identification with induction.* The section map is

   $$f \longmapsto \psi_f, \qquad \psi_f(p) = f(0, h_p).$$

   It carries the induced representation from $A \rtimes \mathrm{SU}(2)$, with inducing data $\chi_{k_m} \otimes D_n$, onto (25.6). Exercise 25.5 checks the intertwining. Then the irreducibility of induced representations in Mackey's theorem, Theorem 25.9(b), shows that the model is irreducible.

**Remark 25.14.** The finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ admit no invariant inner product unless trivial, as proved in Lecture 24. In physics they label the components of fields. They do not classify the particle Hilbert space. For a massive particle of spin $n/2$, that space is $\mathcal{H}_{m,n}$, built from unitary $\mathrm{SU}(2)$ data on an orbit.

## Self-checks

1. Is $(-1,0,0,0)$ in the orbit of $k_1$?
   *Answer.* No. $\Phi(k_1) = I$ is positive definite, but $\Phi(-1,0,0,0) = -I$ is not. Conjugation $\Phi \mapsto g\Phi g^*$ preserves positivity.
2. Do the defining matrices of $H_k$ on $\mathbb{C}^2$ preserve some inner product?
   *Answer.* No. By Proposition 25.11, every finite-dimensional continuous unitary representation of $H_k$ is trivial on $N$. An invariant inner product would make the defining representation such a representation. But $n(1) \neq I$.
3. In the model with $n = 2$, what are $U(z)$ and $W(u,p)$ for $u \in \mathrm{SU}(2)$?
   *Answer.* $U(z) = D_2(-I) = 1$, and $W(u,p) = u$ by the rotation case of the Wigner rotation, Proposition 25.13(e).

## Summary

The Poincaré cover is $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$. Translations act through momentum characters $e^{-ip\cdot a}$, and $\mathrm{SL}(2,\mathbb{C})$ moves momenta by $\Lambda$. The orbits are mass shells, light cones, spacelike hyperboloids and $\{0\}$. Mackey's theorem reduces the classification to little groups. Up to unitary equivalence, irreducible unitary representations with $E$ concentrated on $X_m^+$ are labelled by an $\mathrm{SU}(2)$ spin $n/2$ with $n \geq 0$. Finite-helicity massless representations have trivial little-group translations. Continuous-spin sectors also have positive energy and are separate. The zero-momentum orbit $\{0\}$ gives a further separate sector.

## Reading

- R. Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2: the spectral-measure theorem for vector groups.
- E. P. Wigner, *On Unitary Representations of the Inhomogeneous Lorentz Group*, Annals of Mathematics 40 (1939), 149–204. Read the classification of momentum orbits and little groups.
- A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. Read the treatment of induced representations and Mackey's method for semidirect products.